

Krishna Sai Rohith Vadla

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SUMMARY

Backend and AI engineer who ships LLM infrastructure that keeps running when providers go down. Built a production middleware system that eliminated provider outages as a point of failure across 12 models and 4 providers, an NLP pipeline compressing 35+ live sources at 93% reduction, and a client-facing dashboard running in production at a Wyndham-affiliated property. Strong in Python, FastAPI, and cloud deployment (AWS, GCP); published author at SPECOM 2023.

EDUCATION

Missouri University of Science and Technology

Rolla, MO

Master of Science, Computer and Information Sciences

May 2025

Indian Institute of Information Technology Dharwad

Karnataka, India

Bachelor of Technology, Computer Science and Engineering

May 2023

EXPERIENCE

Data Science Intern

Aug 2025–Present

Neoteksoft Inc.

Dallas, TX

- Cut deployment time from 3 hours to 15 minutes by containerizing GPT-based text generation pipelines with Docker and implementing CI/CD workflows across AWS and GCP.
- Migrated prototype pipeline to production AWS infrastructure with CloudWatch logging and automated model versioning; trained and deployed Transformer and CNN models via HuggingFace on EC2.

Undergraduate Research Assistant

June 2022–May 2023

Indian Institute of Information Technology Dharwad

Karnataka, India

- Co-authored peer-reviewed research (SPECOM 2023, Springer LNAI 14339) on cross-lingual speech style transfer for Soliga, an endangered tribal language; built a 10,000-sentence bilingual corpus and collected ~5,000 studio utterances with a custom recording UI.
- Improved speech intelligibility by 32% over baseline (MOS 3.02 to 3.99, 20 native speakers); reduced generator loss by 23% (32.1 to 24.7) via multi-scale reconstruction loss on a VAE-GAN with WaveNet vocoder.

PROJECTS

LLM Request Router | *FastAPI, PostgreSQL* | Live Demo | GitHub

Dec 2025–Feb 2026

- Reduced failed request rate from 8% to under 1% during provider outages by building a REST API routing layer across 12 LLM models and 4 providers (OpenAI, Anthropic, Google, DeepSeek) with automatic failover, exponential backoff, and database-backed deduplication.
- Designed error classification logic distinguishing transient vs. permanent failures; added request tracing and a real-time dashboard displaying task status, retry counts, and provider health across 4 concurrent API sources.

Hotel Occupancy Dashboard (*Client Deployment*) | *Node.js, PostgreSQL* | Live Demo | GitHub

Aug 2025–Dec 2025

- Eliminated daily 2-hour commute for hotel ownership by replacing manual occupancy tracking with a live dashboard; built automated ingestion pipeline parsing hourly SynXis PMS reports via Gmail IMAP with idempotent processing.
- Currently in production at a Wyndham-affiliated property with signed agreement for expansion to two additional locations; used daily for pricing decisions, staff scheduling, and occupancy monitoring.

Personalized News Summarizer | *HuggingFace Transformers, SQLite* | Live Demo | GitHub

June 2025–Aug 2025

- Reduced article reading time by 93% (500–1000 words → 35–50 words) across 35+ sources using BART summarization and DistilBERT sentiment classification (89% accuracy); deployed on Streamlit Cloud processing 200+ articles daily.
- Added Sentence-BERT personalization layer ranking articles by similarity to user reading history, with sub-2-second page load and zero manual curation.

OPEN SOURCE

Boost Union | *Moodle Theme, 5,000+ installs* | GitHub

Apr 2026

- Contributed OAuth2 button color configuration across `settings.php`, `lib.php`, and `lang/` files; passed PHPUnit and Behat suites, resolved merge conflicts, and delivered production-ready code accepted by maintainers.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++, SQL (PostgreSQL), HTML/CSS

Frameworks: FastAPI, React, Node.js, Flask, Streamlit

AI / ML: Hugging Face Transformers, PyTorch, scikit-learn, BART, DistilBERT, Sentence-BERT, OpenAI / Anthropic / Gemini APIs, multi-provider LLM routing, MLOps

Systems: REST APIs, microservices, distributed task queues, data pipelines, model deployment

Cloud & Tools: AWS (EC2, S3, CloudWatch), GCP, Docker, Git, Railway, MATLAB, VS Code

Libraries: NumPy, pandas, SQLAlchemy, SQLite, Recharts, Plotly, TanStack Query

CERTIFICATIONS

Generative AI Fundamentals — Databricks Academy

June 2025

Python Essentials 1 & 2 — Cisco

Sep 2025

Software Development Virtual Experience — Datacom

Apr 2026